

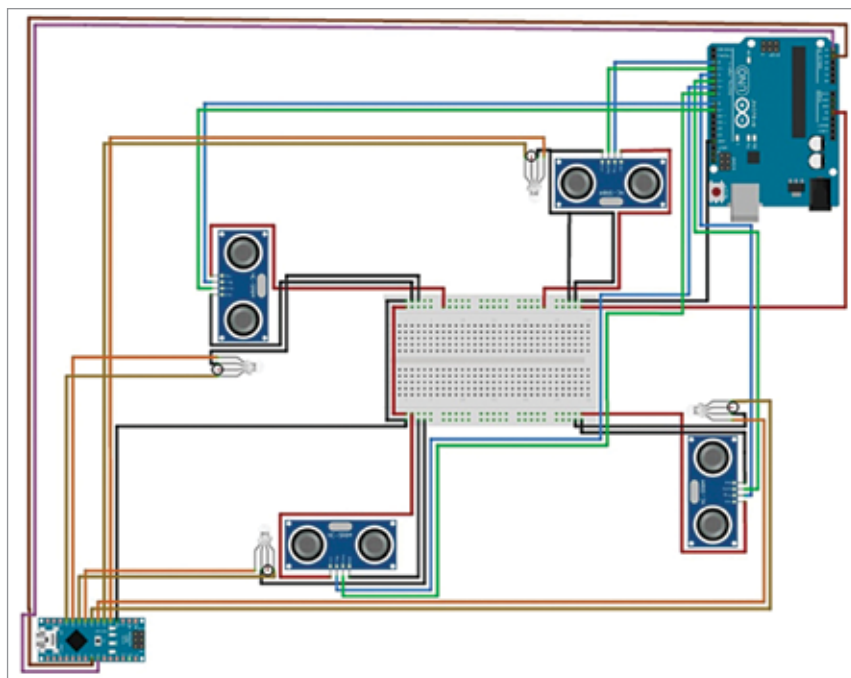


Fig. 6: Wiring diagram of the system

using red jumper wires for 5V, black for GND, and different colours for signal lines. The entire circuit is powered through the USB port. The setup is arranged on a test board to simulate a four-way traffic junction, with each sensor placed at an optimal angle for accurate vehicle detection.

Testing begins by observing serial communication between the two Arduinos after powering on. The Arduino Nano sends real-time distance data from each ultrasonic sensor to the Arduino Uno via the I2C interface. The Uno processes this data to determine which lane should receive the green signal based on detected vehicle density.

When a sensor detects vehicles at a shorter distance, indicating

traffic buildup, the corresponding LED turns green while the others remain red. The green signal duration adjusts according to the number of detected vehicles, ensuring efficient control.

The system is tested under various simulated traffic conditions by placing objects at different distances in front of the sensors to verify response accuracy. Successful testing confirms accurate sensor response, reliable inter-board communication, and correct LED behaviour, validating the performance and reliability of the intelligent traffic control system.

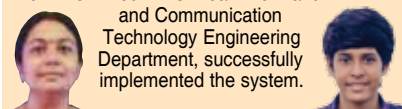
Once powered on, the serial monitor displays vehicle detection and the corresponding traffic light trigger. The signal can also be changed based on vehicle detection using an ultrasonic sensor. **EFY**

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Dr Geetali Saha (left), Faculty of the Department of Electronics and Communication Engineering, GCET, Anand, Gujarat, supervised the work. Prince Tagadiya (right), from the B.Tech First Year Information and Communication Technology Engineering Department, successfully implemented the system.



Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

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